Reg.No.						

Bansilal Ramnath Agarwal Charitable Trust's

VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Examination: ESE

Year: TY **Branch:** Computer

Subject: Design and Analysis of Algorithms **Subject Code:** CS 3205

Max. Marks:60 **Total Pages of Question Paper: 3**

Day & Date: Friday 12 May 2023 **Time:** 11 am to 1 pm

Instructions to Candidate

1. All questions are compulsory.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.

Q. N.	CO No	BT* No		Max marks
Q. 1.				
Q.1. (a)	3	2, 3	For the following two recurrences give best possible asymptotic upper bound on T(n) as a function of n. Express your answer using Big Oh asymptotic notation. (1 mark for correct final answer and 2 marks for the correct approach) (1) T(n) = T(sqrt(n)) +1 where sqrt(n) denotes square root of n. (2) T(n) = 3 T (n/2) + n.	6 marks
Q.1. (b)		3	Give correct recurrence for f(n) for each of the following problems. (1) Let f(n) be different number of binary search trees one can construct using keys {1,2, ..., n} (2) Let f(n) be number of different ways to tile a grid of size 2 x n using rectangular tiles of size 1 x 2 (you can place 1 x 2 tiles vertically or horizontally).	4 marks
Q. 2. (a)	1, 2, 3	1, 3	Let M_1, M_2, ..., M_k be the matrices such that dimension of matrix M_i is **n_i x n_{i+1}** for 1 <= i <= k. For j>=i let f(i,j) denote minimum number of integer multiplications needed to compute the product M_i M_{i+1} ... M_j (assume that we use standard matrix multiplication algorithm to find product of any two matrices). Give recurrence relation for f(i,j) with brief justifying argument of correctness. Specify base cases of the recurrence. Give an efficient dynamic programming based algorithm for computation of f(i,j) using the above recurrence. Specify time and space complexity of your algorithm.	6 marks
Q. 2. (b)			Attempt Any ONE of the following.	4 marks
Q. 2. (b) (i)	4, 6	1	Give prover-verifier based definition of complexity class NP and give two concrete examples of problems characterizing NP class.	
Q. 2. (b) (ii)	5, 6	1	Compare Las Vegas and Monte Carlo randomized algorithms. Give an example for each.	
Q.3.				
Q. 3. (a)	2, 3	2	Consider a text over alphabet {a,b,c,d,e,f,g} relative frequencies of the characters are 5, 7, 8, 16, 18, 22, 24 respectively. Construct Huffman tree. Find Huffman encoding for each of the seven characters. Suppose original text contains total 1,00,000 characters what will be size of the text in bits after encoding. (5 marks for Huffman tree and correct codes, 1 mark for correct text size after encoding)	6 marks

Q. 3. (b)			Attempt any **ONE** of the following	**6 marks**
Q. 3. (b) (i)	1,2,3,6	3, 4	Consider the standard Tower of Hanoi game with n disks. There are 3 pegs A, B, C. There are n disks of distinct sizes placed in Peg A. The disks are numbered from 1 to n, where disk number n is the bottom most disk and disk number 1 is the topmost disk. There are the following two rules of the game. (1) In any move we can move topmost disk in any peg to other peg. (2) We cannot place a larger disk on top of a smaller disk. The goal of the game is to move all disks from Peg A to Peg C with minimum number of moves. Suppose you are given natural numbers n, N, i as input where N<= 2^n -1 and 1 <= i <= n. You want to figure out how many times disk number i gets moved in the first N moves of the tower of Hanoi game with n disks. Give an efficient algorithm for this task. Give proper Pseudo-code and analyze time complexity of your algorithm. For full credit, time complexity of your algorithm (as a function of n) must be asymptotically equal to the time complexity of the best algorithm for the problem.	
Q. 3. (b) (ii)	1, 2, 3, 6	3, 4	Let S be set of n integers each of which is less than k and greater than -k. You are given set S and number b, such that -nk < b <nk and goal is to check if there is a subset of S such that the addition of all the elements in the subset is b. Give an algorithm which has time complexity bounded above by O((nk)^c) for an absolute constant c. Give detailed pseudo-code and the complexity analysis. (Hint: Note similarity with coin change problem)	
Q.4.	1, 2,3 5,6	4, 6	Given integers n and k, and numbers p_1, … , p_n ∈ [0, 1] (that is 0 <= p_i <= 1 for every i). You want to determine the probability of obtaining exactly k heads when n biased coins are tossed independently at random, where p_i is the probability that the ith coin comes up heads. Give an O(n^2) time algorithm for this task with proper Pseudo code. Give an argument justifying correctness of the algorithm and the time complexity. Assume you can multiply and add two numbers in [0, 1] in O(1) time.	**8 marks**
	1, 2, 3, 6	4, 6	Suppose n circles are placed in 2-D plane. You are given coordinates of center and radius for each of the circles. A ray is a line segment starting at origin (0,0) and extending all the way up-to infinity in any given direction. Give an efficient algorithm to find the minimum number of rays needed to cover all the circles. We say a ray covers a circle if the ray passes through the circle non-trivially or the ray is tangential to the circle. Prove the correctness of your algorithm. Assume that you can do any geometric task such as finding intersection point(s) of a line and a circle, finding a point of contact of a circle and a tangent drawn from point outside the circle etc in O(1) time. (3 marks for the correct algorithm, 5 marks for the correctness proof of the algorithm) For full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	**8 marks**
Q.6.				
Q. 6. (a)	1, 2, 3, 6	4, 6	You are given two **sorted** arrays A and B each containing n distinct integers and a natural number r <= 2n as input. Assume that there is no common element in A and B. Let C be set of all elements either in A or B (that is C is union of A and B). For an element p in array C rank(p) in C is a total number of elements in C which are less than or equal to p. (E.g. if C = {2, 7, 1, 9} then rank(7) in C is 3 as elements 1, 2, 7 in C are less than or equal to 7) . Give an efficient algorithm to find an element p in C with rank(p) = r. Give proper Pseudo-code for the algorithm. Analyze the time complexity of the algorithm. For full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	**7 marks**
Q. 6. (b)	1, 2, 3	3, 4	You are given directed acyclic graph G(V,E) and vertices s, t in V as input. Give an efficient algorithm to compute number of distinct directed paths from vertex s to vertex t in G. Give proper Pseudocode of the algorithm and analyze its time complexity. For full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	**5 marks**